Lesson Objective

Plot data on line graphs and analyze trends.

Materials

(S) Mini-Lesson Worksheet, personal white board

PROBLEM

1 **Plot given data points on a line graph and connect them to show a pet's weight over time.**

The table below shows Biscuit the dog's weight at different ages.

Age (months)	Weight (pounds)
0	3
4	15
8	25
12	35
16	40
20	45
24	45
28	45

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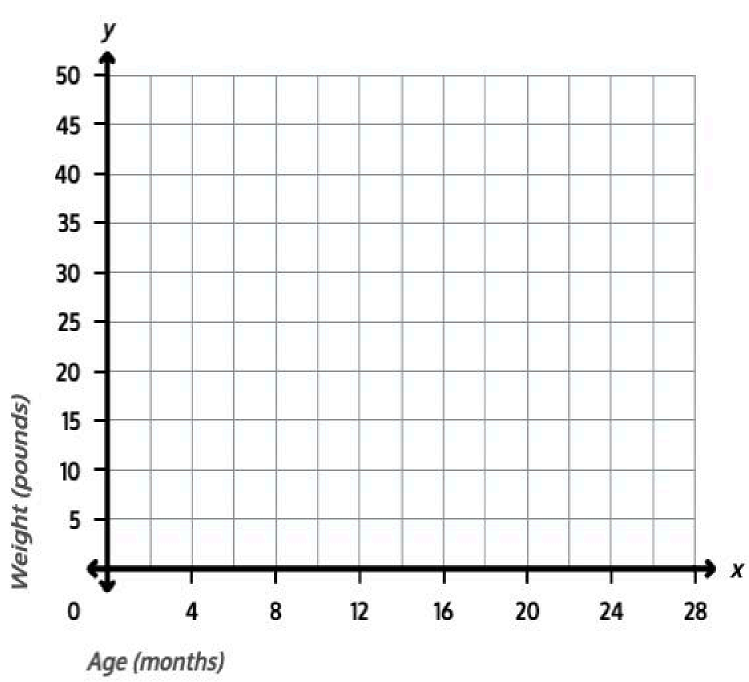
Plot each data point on the coordinate plane below. Then, connect the points with line segments to make a line graph.



TEACHER GUIDANCE

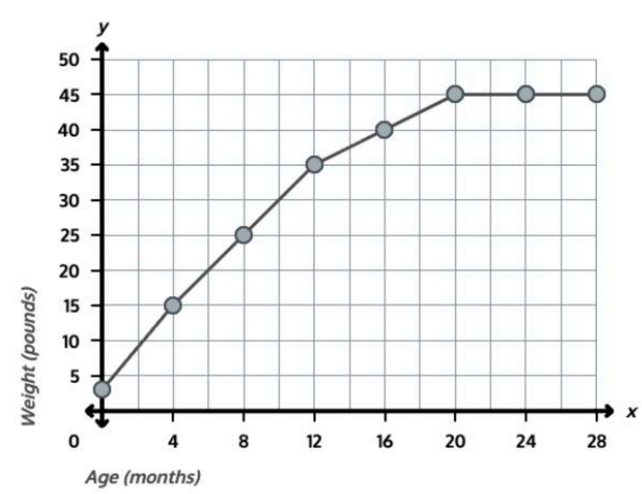
Guide students through reading the data table and plotting points on the coordinate plane.

Tell students that the x-axis shows Biscuit's age in months and the y-axis shows his weight in pounds.

Model plotting the first point: at 0 months, Biscuit weighed 3 pounds, so plot the point at (0, 3). Model plotting the second point: at 4 months, Biscuit weighed 15 pounds, so move to 4 on the x-axis and up to 15 pounds on the y-axis.

Have students plot the remaining points from the table.

Tell students to connect the points with line segments to complete the line graph.



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PROBLEM	TEACHER GUIDANCE

Answer: Points plotted at (0, 3), (4, 15), (8, 25), (12, 35), (16, 40), (20, 45), (24, 45), (28, 45) and connected with line segments.

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2 **Use the line graph to find how much weight was gained during different time periods by subtracting coordinates.**

Use the line graph of Biscuit's weight to answer the following questions.

a. a. How much weight did Biscuit gain in the first 4 months of his life, between 0 months and 4 months?
Answer: 12 pounds

b. How much weight did Biscuit gain in the second 4 months of his life, between 4 months and 8 months?
Answer: 10 pounds

c. How much more weight did Biscuit gain in the first 4 months than in the second 4 months?
Answer: 2 pounds

Guide students through part (a). Ask students to find Biscuit's weight at 0 months and at 4 months on the line graph.

Ask students how they can find how much weight Biscuit gained during that time. Tell students to subtract: 15 pounds − 3 pounds = 12 pounds. Point out that the difference in the y-coordinates tells us how much weight Biscuit gained.

Have students solve part (b) using the same strategy. Ask students to find Biscuit's weight at 4 months and at 8 months, and subtract: 25 pounds − 15 pounds = 10 pounds.

Have students solve part (c). Ask students to compare the two amounts: 12 pounds − 10 pounds = 2 pounds. Point out that Biscuit gained 2 more pounds in the first 4 months than in the second 4 months.

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3 **Analyze the line graph to identify when growth was fastest, slowest, or unchanged by comparing the steepness of line segments.**

Use the line graph of Biscuit's weight to answer the following questions.

a. During which 4-month period did Biscuit gain weight the fastest? How can you tell from the graph?
Biscuit gained weight the fastest between 0 months and 4 months because that line segment is the steepest

b. What happened to Biscuit's weight between the ages of 20 months and 28 months? How does the line graph show this?
Biscuit's weight stayed the same at 45 pounds. The line graph shows a horizontal (flat) line during this time, meaning there was no change in weight.

Students work independently.

Ask students to look across the whole line graph and find the steepest line segment for part (a), and to describe what the flat line means for part (b).

Monitor For

- Do students accurately locate each data point by finding the correct x-value first and then moving up to the correct y-value on the coordinate plane?
- Do students subtract the y-coordinates in the correct order (later weight minus earlier weight) to find the amount of weight gained?
- Do students connect steepness of a line segment to how fast growth occurred, rather than just looking at the overall height of the line?
- Can students explain that a horizontal line segment means no change in weight, not just that the weight is high?

Instructional Moves

- If a student plots a point at the wrong location, have them point to the age on the x-axis first, trace straight up, then find the weight on the y-axis and trace across. The point goes where those paths meet.
- If a student confuses "how much weight gained" with "total weight," remind them to subtract: weight gained = weight at the end minus weight at the start of the period.
- When comparing line segments in Problem 3, ask students which segment rises more from left to right. The segment that rises the most over the same horizontal distance is the steepest, meaning growth was fastest.
- If a student says the flat line means Biscuit stopped growing "because the line stopped," redirect by asking what the y-coordinates are at 20 months, 24 months, and 28 months — all are 45 pounds, so the weight stayed the same.

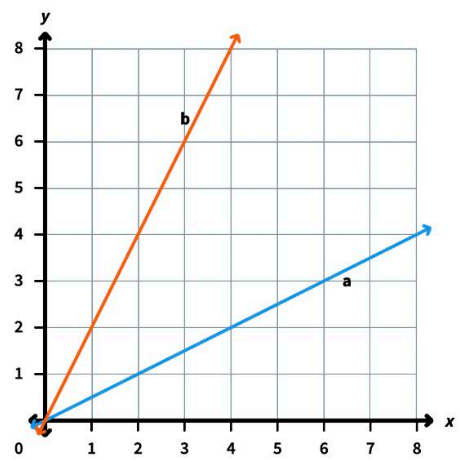
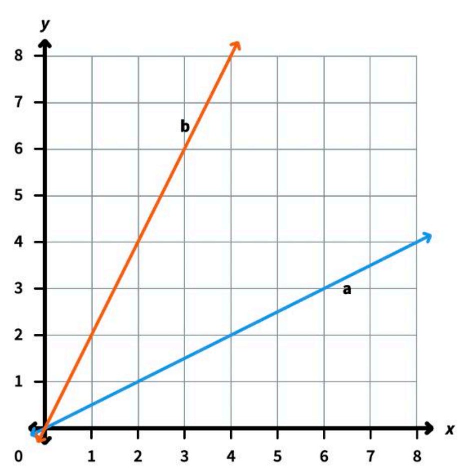
Reflection Questions

1. How did you find how much weight Biscuit gained during a 4-month period using the line graph?
Answer: I found Biscuit's weight at the start and end of the period on the y-axis and subtracted to find the difference.
2. How can you tell from looking at the line graph which 4-month period had the fastest weight gain?
Answer: The steepest line segment shows the fastest weight gain. The segment from 0 months to 4 months is the steepest, so that period had the fastest gain.
3. What does a flat, horizontal line on a line graph tell you about the data?
Answer: A flat line means the value did not change during that time. Biscuit's weight stayed at 45 pounds between 20 months and 28 months.

Mini Lesson Sample Script

Materials

(S) Mini-Lesson Worksheet, personal white board

PROBLEM	SAMPLE SCRIPT
1 **The table below shows Biscuit the dog's weight at different ages.**  Plot each data point on the coordinate plane below. Then, connect the points with line segments to make a line graph.  *Answer: Points plotted at (0, 3), (4, 15), (8, 25), (12, 35), (16, 40), (20, 45), (24, 45), (28, 45) and connected with line segments.*	**T**: Here's a table showing how much a dog named Biscuit weighed at different ages. We're going to use this data to build a line graph. Take a look at the coordinate plane. What does the x-axis show? **S**: Biscuit's age in months. **T**: And the y-axis? **S**: His weight in pounds. **T**: Right. Let's plot the first point together. At 0 months, Biscuit weighed 3 pounds. So I start at 0 on the x-axis and go up to 3 pounds on the y-axis. That's the point (0, 3). Plot it on your graph. **S**: (Plot.) **T**: At 4 months, Biscuit weighed 15 pounds. I move to 4 on the x-axis and go up to 15 pounds. Plot that one too. **S**: (Plot.) **T**: Good. Now use the table to plot the rest of the points on your own. **S**: (Plot remaining points.) **T**: Now connect all the points with line segments, starting from the first point to the second, the second to the third, and so on. **S**: (Connect the points.) **T**: Now we have a line graph that shows how Biscuit's weight changed over time.

PROBLEM	SAMPLE SCRIPT

2 **Use the line graph to find how much weight was gained during different time periods by subtracting coordinates.**

Use the line graph of Biscuit's weight to answer the following questions.

a. How much weight did Biscuit gain in the first 4 months of his life, between 0 months and 4 months?

b. How much weight did Biscuit gain in the second 4 months of his life, between 4 months and 8 months?

c. How much more weight did Biscuit gain in the first 4 months than in the second 4 months?

Answer: a) 12 pounds; b) 10 pounds; c) 2 pounds

T: Look at the line graph. What did Biscuit weigh at 0 months?

S: 3 pounds.

T: And at 4 months?

S: 15 pounds.

T: So how can we figure out how much weight he gained in those first 4 months?

S: Subtract. 15 pounds minus 3 pounds is 12 pounds.

T: So Biscuit gained 12 pounds in the first 4 months. The difference in the y-coordinates tells us how much weight he gained. Now try part (b). How much weight did Biscuit gain between 4 months and 8 months?

S: At 4 months he weighed 15 pounds, and at 8 months he weighed 25 pounds. 25 pounds minus 15 pounds is 10 pounds.

T: So he gained 10 pounds. Now for part (c) — how much more weight did Biscuit gain in the first 4 months than in the second 4 months?

S: 12 pounds minus 10 pounds is 2 pounds. He gained 2 more pounds in the first 4 months.

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3 **Analyze the line graph to identify when growth was fastest, slowest, or unchanged by comparing the steepness of line segments.**

Use the line graph of Biscuit's weight to answer the following questions.

a. During which 4-month period did Biscuit gain weight the fastest? How can you tell from the graph?

b. What happened to Biscuit's weight between the ages of 20 months and 28 months?

T: Look across the whole line graph and answer parts (a) and (b) on your own.

S: (Work.)

PROBLEM	SAMPLE SCRIPT
How does the line graph show this? *Answer: a) Biscuit gained weight the fastest between 0 months and 4 months because that line segment is the steepest; b) Biscuit's weight*	